

MAE 3128, Spring 2026

Homework assignment 3

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This assignment is due at the end of the day on Wednesday, April 8, 2026. Collaboration is encouraged. Solutions will be graded on the following criteria: Correct answer, correct units, appropriate significant digits, problem statement and assumptions, and neatness.

HW3 P1: Find $\sigma_{z\theta}$, $\sigma'_{z\theta}$, and $\sigma'_{zz}|_{\max}$ given a positive torque T_o applied at the free end of a constant-diameter solid shaft of radius c and length L and having a shear modulus G . Assume that the shaft is fixed at the left end.

$$\sigma_{z\theta}^{(r)} = \frac{\tau r}{J} \quad J = \int r^2 dA = \frac{\pi c^4}{2}$$

$$\sigma'_{z\theta} = \frac{\tau r}{J} (\cos^2 \alpha - \sin^2 \alpha)$$

In pure torsion: $\sigma'_{zz}|_{\max} = \sigma'_{z\theta}(c) \Rightarrow \tau_{\theta z}(r) = \frac{T_o r}{\pi c^4/2}$

$$\sigma'_{z\theta} = \frac{2T_o r}{\pi c^4} (\cos^2 \alpha - \sin^2 \alpha)$$

$$\Rightarrow \sigma'_{z\theta} = \frac{2T_o r}{\pi c^4}$$

$$\sigma'_{zz}|_{\max} = \sigma'_{z\theta}(c) = \frac{2T_o c}{\pi c^4}$$

$$\Rightarrow \sigma'_{zz}|_{\max} = \frac{2T}{\pi c^3}$$

HW3 P2: In 1984, Lanyon and Rubin suggested that the number of cycles of loading per day and the maximum achieved strain both serve as mechanobiological stimuli for bone growth. In particular, they found that bone mass was maintained (i.e., production and removal were balanced) given a strain history of 4 cycles/day at a maximum value of 0.002 or, similarly, at 100 cycles/day at a maximum value between 0.0005 and 0.001 (assume 0.0008). They suggest that these combined effects can be accounted for via a "daily bone stimulus" parameter ξ that is computed via the following formula

$$\xi = \left(\sum_{\text{day}} n \epsilon^m \right)^{1/m}$$

where n is the number of cycles/day, ϵ is the maximum strain attained per cycle, and m is an empirically determined material parameter. Given the data listed here, find a value for m .

$$\left[\sum_{\text{day}} 4 (0.002)^m \right]^{1/m} = \left[\sum_{\text{day}} 100 (0.0008)^m \right]^{1/m}$$

$$4 (0.002)^m = 100 (0.0008)^m$$

$$\frac{1}{25} = \left(\frac{0.0008}{0.002} \right)^m = 0.4^m$$

$$\log\left(\frac{1}{25}\right) = m \log(0.4)$$

$$\Rightarrow m = \frac{\log(1/25)}{\log(0.4)}$$

$$\Rightarrow m = 3.513$$

HW3 P3: Based on the results of HW3 P2, determine the number of cycles that one should walk per day if the strain during normal walking is $400 \mu\epsilon$ (i.e. microstrain, where $1 \mu\epsilon = 1 \times 10^{-6}$).

$$[4(0.002)^m]^{1/m} = [n(400 \times 10^{-6})^m]^{1/m}$$

$$n = \frac{4(0.002)^m}{(400 \times 10^{-6})^m}$$

Recall:
 $m = 3.513$ { from HW3 P2 }

$$\Rightarrow n = \frac{4(0.002)^{3.513}}{(400 \times 10^{-6})^{3.513}}$$

$$\Rightarrow \boxed{n = 1142 \text{ cycles/day}}$$

HW3 P4: If a 17.2 Nm torque induces a maximum shearing strain of $1,132 \mu\epsilon$ at the periosteal surface in the diaphysial region of the femur, what is the associated value of the shearing stress if the shear modulus is 3.3 GPa?

Given: $G = 3.3 \text{ GPa} = 3.3 \times 10^9 \text{ Pa}$; $\gamma = 1,132 \mu\epsilon = 1.132 \times 10^{-6}$

$$\tau_{20} = (3.3 \times 10^9 \text{ Pa})(1.132 \times 10^{-6})$$

$$\tau_{20} = 2G\epsilon_{20}$$

$$\Rightarrow \tau_{20} = (3.3 \times 10^3)(1.132) \text{ Pa}$$

$$\boxed{\tau_{20} = G\gamma}$$

$$\left\{ \because \gamma = 2\epsilon_{20} \right\}$$

$$\boxed{\begin{aligned} \tau_{20} &= 3.735 \times 10^6 \text{ Pa} \\ &= 3.735 \text{ MPa} \end{aligned}}$$

HW3 P5: A long bone is subjected to a torsion test. Assume that the inner diameter is 0.375 in. and the outer diameter is 1.25 in., both for a circular cross section. If $E = 16 \text{ GPa}$ and $\nu = 0.325$, find the largest torque that can be applied prior to yield, where $\sigma_{\text{yield}} = 1.25 \text{ ksi}$ (i.e. a maximum normal stress).

$$\tau_{2\theta} = \tau_1 = \frac{Tc}{J}$$

$$J = \frac{\pi(C^4 - r_i^4)}{2}$$

Given:

$$T_o = C = 0.625 \text{ in} = \frac{1.25 \text{ in}}{2}$$

$$r_i = 0.1875 \text{ in} = \frac{0.375 \text{ in}}{2}$$

$$E = 16 \text{ GPa} = 16 \times 10^9 \text{ Pa}$$

$$\nu = 0.325$$

$$\sigma_{\text{yield}} = 1.25 \text{ ksi} = 1.25 \times 10^3 \text{ psi}$$

$$J = \frac{\pi}{2} (0.625^4 - 0.1875^4) \text{ in}^4 = 0.238 \text{ in}^4$$

$$T = \frac{\sigma_{\text{yield}} J}{C} = \frac{(1.25 \times 10^3 \text{ lb/in}^2) (0.238 \text{ in}^4)}{0.625 \text{ in}}$$

$$T \approx 475.5 \text{ lb.in} \approx 53.72 \text{ Nm}$$